

Machine Learning with Python Project Report

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Project Summary

The Department of Roads and Transportation is grappling with the escalating concern of road accidents. To address this issue, the department has initiated a project to leverage emerging technology for predicting the probability of road accidents. This report details the creation of a toy dataset, the implementation of machine learning models, and the development of a predictive tool based on user input.

User Interaction and Prediction

The developed tool allows users to input information about gender, age, alcohol consumption, fatigue, time of day, and weather conditions. The logistic regression model then predicts the probability of a road accident based on this input. The probability estimates are provided to users, offering insights into the potential risk of an accident.

Choice of Model

In the provided scenario we have been contracted to provide three machine learning models that are capable of predicting the probability of a road accident, based on a variety of factors at the time of the accident. This is a binary classification problem, as it requires us to classify whether or not there will be a road accident, so a categorical supervised machine learning model was chosen in order to predict target values based on multiple features and return corresponding probabilities. Categorical supervised machine learning models are used for classification tasks where the target variable has discrete categories or classes. To get probability estimates for class membership, you can use models that provide probability estimates for each class. However, there are various categorical supervised machine learning models that can return class probabilities. Logistic Regression, Naive Bayes, Random Forest, Gradient Boosting, Support Vector Machines (SVM) and K-Nearest Neighbors. When using any of these models, the `predict_proba()` function can be used to get class probabilities. To decide which one to use, all of them were run to see the difference in accuracy score. This was done in the now commented code below :

```
random_seed = 40
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=random_seed)

clf1 = LogisticRegression()
clf1.fit(x_train, y_train)
print(f'Log Regression Accuracy: {clf1.score(x_test, y_test)}')

clf2 = DecisionTreeClassifier()
clf2.fit(x_train, y_train)
print(f'Tree Accuracy: {clf2.score(x_test, y_test)}')

clf3 = RandomForestClassifier()
clf3.fit(x_train, y_train)
print(f'Random Forrest Accuracy: {clf3.score(x_test, y_test)}')

clf4 = SVC()
clf4.fit(x_train, y_train)
print(f'SVC Accuracy: {clf4.score(x_test, y_test)}')

clf5 = CategoricalNB()
clf5.fit(x_train, y_train)
```

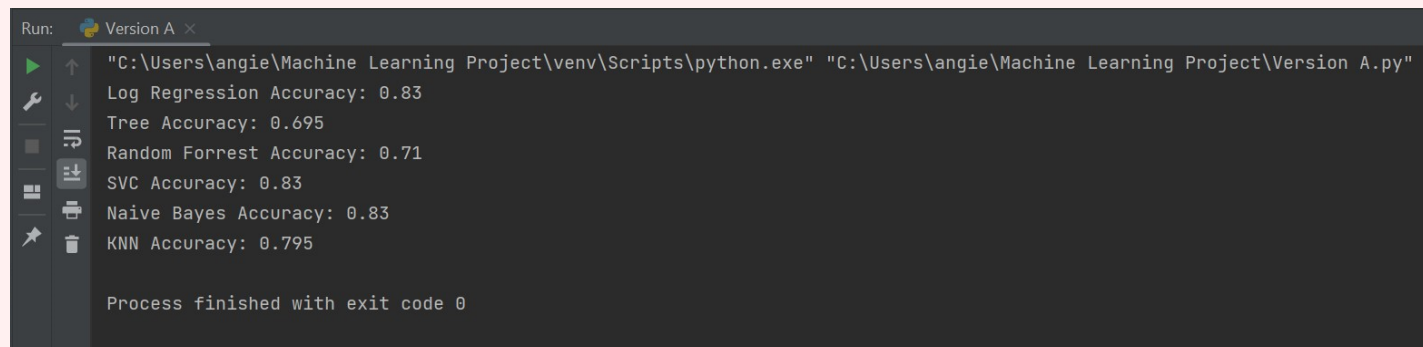
```
print(f'Naive Bayes Accuracy: {clf5.score(x_test, y_test)}')
```

```
clf6 = KNeighborsClassifier()
```

```
clf6.fit(x_train, y_train)
```

```
print(f'KNN Accuracy: {clf6.score(x_test, y_test)}')
```

Result:



The screenshot shows a terminal window titled 'Run: Version A'. It displays the output of a Python script that evaluates the accuracy of several machine learning models. The results are as follows:

Model	Accuracy
Log Regression	0.83
Tree	0.695
Random Forrest	0.71
SVC	0.83
Naive Bayes	0.83
KNN	0.795

The terminal also shows the command used to run the script: `"C:\Users\angie\Machine Learning Project\venv\Scripts\python.exe" "C:\Users\angie\Machine Learning Project\Version A.py"` and a final message: `Process finished with exit code 0`.

The Logarithmic Regression, Support Vector Machines and Naive Bayes models scored the best, so we chose one of these: Logarithmic Regression.

Future Directions

- **Model Comparison:** Further evaluate and compare the performance of other classifiers (Decision Tree, Random Forest, SVC, Naive Bayes, KNN) to select the most suitable model for the task.
- **Feature Engineering:** Investigate the inclusion of additional features or engineering new features to enhance model performance
- **Hyperparameter Tuning:** Optimize the hyperparameters of selected models to achieve better predictive accuracy.
- **Real Data Integration:** Incorporate real-world accident data for model training to improve the model's ability to generalize to actual scenarios.
- **User Interface Enhancement:** Consider developing a more interactive and user-friendly interface for wider adoption.

Conclusion

The road accident prediction project is an essential step toward proactively addressing the issue of accidents. The logistic regression model demonstrates promising accuracy in predicting accidents based on various factors. Further exploration and comparison of alternative machine learning models are recommended to ensure the selection of the most robust predictive tool.

The tool provides a user-friendly interface for obtaining real-time accident probability estimates, aiding both the Department of Roads and Transportation and individual users in making informed decisions. As the project evolves, additional features, model refinements, and data enhancements can be explored for continuous improvement. In conclusion, the road accident prediction project is a significant step toward leveraging technology for safer roads. Continued collaboration with stakeholders, ongoing model refinement, and the incorporation of real-world data will contribute to the success and impact of this initiative.